AI for SMEs - Handbook

A living handbook that helps regional manufacturing SMEs put AI to work — written by HAN students, one practical question at a time.

This handbook is written for owners and managers of small and medium-sized home-care organisations who are new to AI and do not need to be AI experts. It helps understand how AI-supported planning or notifications could help and what must be checked before choosing a tool.

The handbook does not replace professional judgement and does not guarantee that a specific AI tool will work. It also does not prove that Thuiszorg Totaal has the same problem as a larger provider. The original observation came from STMG and is treated only as a hypothesis to test at Thuiszorg Totaal.

The questions this handbook answers

The handbook is organised by themes, and every theme is posed as a question an owner would actually ask — not as a topic. The themes are the stable structure; the chapters beneath them compound from cohort to cohort. One theme, one team, one chapter.

#	Theme	The SME-facing question
1	Geopolitics and vendor models	Who owns this thing and what happens to our organisation if that changes?
2	Trust and AI Safety	Can I trust this, with my data, with its answers and with my people?
3	SME knowledge gaps	What does Thuiszorg Totaal not yet know about AI, and what would it need to know to decide well?
4	AI use cases at SMEs	What do firms like ours use AI for, and what pays off?

Note on numbering: an earlier draft of this handbook skipped from Theme 2 straight to "Theme 4," with no Theme 3 ever written — nothing was dropped, the team confirmed no separate chapter was produced for that slot. The chapters below are renumbered 1–4 with no gap.

Theme 1 — Who owns this thing and what happens to our organisation if that changes?

This chapter helps a Dutch home-care SME understand why it matters who owns and operates the AI tool, where the data goes, and what happens if the vendor or its technology changes.

Team 3 · version 3 · SME: Thuiszorg Totaal · sources last appraised 16 September 2026 · reason for this version: added a cross-reference to Theme 4's client-notification finding

The question, for this kind of SME. If Thuiszorg Totaal uses an AI-supported scheduling or notification tool, does it matter whether the tool comes from a US hyperscaler, a European vendor or a Dutch care-sector supplier, and what happens if that vendor changes?

The translation. What matters the most for Thuiszorg Totaal when choosing an AI tool for scheduling/notifications is the full vendor chain; who provides the software, where the data is stored, which cloud or AI providers are involved, and what happens if one of these companies changes. This is especially important in home care because the system may process sensitive personal or health data. Under GDPR (general data protection regulation), Thuiszorg Totaal remains responsible for this data even when an external AI provider processes it. (AP Publiceert Praktijkgids 'Gezondheidsgegevens in de Cloud', z.d.)

A US vendor is not automatically unsafe, and a Dutch or European vendor is not automatically safe. A Dutch supplier may still use an American cloud or AI provider behind its service. For data transfers outside the European Economic Area (EEA), Thuiszorg Totaal must check if a valid GDPR transfer mechanism applies. (EU-US Data Transfers, 2020)

What happens if the vendor changes? A vendor may be bought by another company, change its cloud or AI provider, add new sub-processors, or move data processing to another country. This can change who has access to Thuiszorg Totaal's data and which rules apply. This does not have to mean the service must stop. Thuiszorg Totaal should be informed about important changes and check whether the new situation still meets GDPR requirements. If the change is not acceptable, the company should be able to object, switch providers, and export or delete its data.

Illustrative use case (CRISP-DM). Example: Thuiszorg Totaal wants to test an AI-supported tool that helps planners create schedules and notify clients about expected care visits.

(Theme 4's own evidence search later found no documented adopter case for automatic client/family notifications specifically — only vendor claims. Treat the "notify clients" part of this illustration as a generic example of the vendor-chain question, not as a use case with evidence behind it yet; see Theme 4, section 2.)

- **Business understanding:** The goal is to reduce manual planning and communication work while keeping the planner in control.
- **Data understanding:** First, Thuiszorg Totaal identifies what data the tool would need, such as employee availability, client addresses and appointment times. If health information is involved, this requires extra protection under the GDPR.
- **Data preparation:** Only the data that is actually needed should be shared. Thuiszorg Totaal also checks where the data will be stored and which cloud, AI and other providers will have access to it.
- **Modelling:** The AI can use the available planning data to suggest schedules or help organise changes. A human planner reviews and approves important decisions.
- **Evaluation:** During a small pilot, Thuiszorg Totaal checks whether the tool actually saves time, produces useful results and handles data as agreed. Problems and incorrect suggestions are recorded.
- **Deployment:** If the pilot works, the tool can be introduced gradually. Before wider use, Thuiszorg Totaal should have clear agreements about data access, vendor changes, data export and deletion, so it can remain in control or switch providers if necessary.

Recommended tooling. Thuiszorg Totaal should start with the simplest option before buying/integrating a complex AI system.

Option	What it could do	Effort
No-code/low-code platform	Build the first prototype and connect scheduling information without developing a complete system from scratch	Low-medium: requires setup, testing, and basic training

Option	What it could do	Effort
AI scheduling component	Use planning data such as employee availability, locations and time constraints to suggest schedules	Medium-high: requires suitable data, testing and human review
Notification integration	Automatically send approved schedule updates to clients or employees through SMS, email or an app	Medium-high: requires connection to the scheduling system and testing or building an app

Responsible AI & trust. Thuiszorg Totaal remains responsible for its personal and health data when an external provider processes it. The company should therefore know what data is shared, who can access it, where it is processed and how it can be deleted or returned. This follows the guidance of the Dutch Autoriteit Persoonsgegevens for organisations using cloud services for health data. AI output should also not be trusted automatically. During a pilot, planners should check suggested schedules, record mistakes and confirm important changes themselves. The AI should support the planner, not replace human control. For employees, this means being informed about how the tool is used and having a clear way to correct incorrect schedules or decisions. Only the data needed for the task should be shared with the system.

First five steps for an owner:

1. **Map the problem:** Identify which scheduling or notification task actually needs improvement.
2. **Map the data:** Write down what client and employee data the tool would need.
3. **Check the vendor:** Ask who owns the tool, where the data is processed and which cloud, AI and sub-processors are involved.
4. **Run a small pilot:** Test the tool with limited data and let planners check its output.
5. **Plan the exit:** Before committing, make sure Thuiszorg Totaal can export and delete its data and switch providers if the vendor or conditions change.

Sources.

- AP publiceert praktijkgids 'Gezondheidsgegevens in de cloud'. (z.d.). Autoriteit Persoonsgegevens. autoriteitpersoonsgegevens.nl/actueel/ap-publiceert-praktijkgids-gezondheidsgegevens-in-de-cloud
- Data controller or data processor | Data protection guide for small business | European Data Protection Board. (z.d.). edpb.europa.eu/sme/learn-the-basics/data-controller-or-data-processor_en
- EU-US data transfers. (2020, 10 augustus). Europäische Kommission. commission.europa.eu/law/law-topic/data-protection/international-dimension-data-protection/eu-us-data-transfers_de
- Nadibaidze, A., & Lingevicius, J. (2026, maart). Geopolitics of AI Governance Actors, Norms, and Trends. findresearcher.sdu.dk
- Rules on international data transfers. (z.d.). European Commission.
- Sánchez, E., Calderón, R., & Herrera, F. (2025, 9 juni). Artificial Intelligence Adoption in SMES: Survey based on TOE–DOI Framework, Primary Methodology and challenges. mdpi.com

Educational, not consultancy. This handbook offers inspiration and starting points, not commercial advice or guarantees. Decisions about a firm remain the firm's own.

Theme 2 - Can I trust this, with my data, with its answers and with my people?

This chapter helps the owner of a small Dutch home-care organisation decide what to hand to an AI scheduling or notification tool, what to keep in the building, and what to ask staff who may already be using AI on their own.

Team 3 · version 2 · SME: Thuiszorg Totaal · sources last appraised 23 September 2026 · reason for this version: added a cross-reference to Theme 4's client-notification finding

1. The question, for this kind of SME.

Chapter 1 asked who owns the tool and where the data goes. This chapter assumes that check is done and covers the next question. If Thuiszorg Totaal uses an AI tool to plan schedules and notify clients, can it trust the tool with client and employee data, with its suggestions, and with its own planners and carers? The size of Thuiszorg Totaal and how much AI its staff already use are unknown. As in Chapter 1, they are hypotheses to test.

2. The translation. Could the firm run the model itself?

There are three places an AI can run, and your data travels further with each step out.

- **A free public chatbot.** The Dutch privacy regulator has received breach notifications where staff typed personal data, for example of patients, into an AI chatbot. The provider can then gain access to that data, and if the employee did it against the employer's rules it counts as a data breach (Autoriteit Persoonsgegevens, 2024). A December 2025 incident at the municipality of Eindhoven showed the scale: a 30-day sample found CVs, youth-care documents and internal reports in public chatbots (Binnenlands Bestuur, 2025).
- **A business tool run by a provider.** Your input passes through the provider's systems, and providers may log inputs and outputs. The EDPB expert report advises users to limit sensitive data and to make sure the provider does not keep or misuse it (Barberá, 2025). Chapter 1 covers the vendor chain.
- **A model on your own hardware ("open-weight").** This keeps data in your own environment, but needs hardware, technical expertise and maintenance, and breaches remain possible because security depends on your whole IT set-up (Dennstädt et al., 2025). The same authors note that a hosted open model, run by a provider, is a middle route. For a small firm, running its own model is a later option, not a starting point.

Whether data is sensitive depends on the task, and pseudonymised data can rarely be made fully anonymous in a care setting (Dennstädt et al., 2025). The starting rule below is team judgement, not a sourced finding, and must be reviewed by a privacy officer.

Hand to the AI (after the vendor check and a contract)	Keep internal
Employee availability windows and qualification levels	Care notes, diagnoses, medication, care plans
Visit time slots and travel times (calculate them in your own system and send only the result)	Citizen service number (BSN) and free-text messages from clients
Client codes instead of names	Anything linking a named client to the reason for care
Fixed-template reminder texts for visits a planner has already approved	Employee HR data: sickness reasons, performance, disciplinary matters

2.1 When may the AI act for the firm, and when only advise?

We propose three levels. The evidence below explains why the middle level is the default.

Level	Example at Thuiszorg Totaal	Rule
Acts alone (after a successful pilot)	Sends a reminder for a visit the planner already approved, from a fixed template	Log every message and make sure the client can always reach a person
Suggests, planner decides	Drafts a weekly schedule, proposes a replacement when a carer calls in sick and drafts messages about changes	Planner approves each change and overrides are logged with a reason

Level	Example at Thuiszorg Totaal	Rule
Never alone	Anything that changes a carer's hours, pay or assessment, anything about care content or urgency and replies to a client's health question	A person decides and is named

Three findings shape these levels.

6. **"A human approves" only counts if it is real.** In the SCHUFA judgment, the EU Court of Justice ruled that a score can itself be an automated decision when others rely heavily on it (Aza, 2024). Legal analysis of that ruling concludes that what matters is whether the final decision rests decisively on the automated output, even when a person formally decides (Laux & Ruschemeier, 2025). A planner who approves everything in seconds provides no protection. Overreliance on AI output ("automation bias") is well documented and is named in the AI Act itself (Laux & Ruschemeier, 2025).
7. **Regulators have fined algorithmic shift management.** Italy's data protection authority fined delivery platform Foodinho €5 million in 2024 for, among other things, automated processing of couriers' data without any way for couriers to get human intervention or contest decisions (Eurofound, 2024). This is a different sector and scale. We use it as an illustration, not a comparison.
8. **Scheduling tools can fall in the AI Act's high-risk list.** The Act lists systems that allocate tasks to workers based on their behaviour or personal traits (European Commission, AI Act Service Desk, Annex III). Whether a given tool does so depends on what it uses to allocate visits, so ask the vendor to state its class in writing. For such systems, the employer must assign oversight to someone with competence, training and authority, and must inform workers and their representatives before use (European Commission, AI Act Service Desk, Article 26). The high-risk obligations were postponed from August 2026 to 2 December 2027 by the Digital Omnibus (DLA Piper, 2026; Cooley, 2026). Plan for them anyway.

2.2 What happens when staff use AI and do not say so?

Assume it is happening. A survey of more than 48,000 people in 47 countries found that 57% of employees hide their AI use and present AI-generated work as their own, and almost half admit to uploading sensitive company information into free public tools. Only 47% had received AI training and only 40% said their workplace had a policy (Gillespie, Lockey et al., 2025). Half feared being left behind, which suggests people hide AI use out of pressure rather than bad intent. The survey is global and self-reported, so Dutch home care may differ.

Why does it matter legally? If client data was typed into a public chatbot, it is a data breach that must be reported to the regulator within 72 hours of the firm learning of it, and it can also breach medical professional secrecy (AKD, n.d.). Decide what you will do with a "yes" before you ask.

What to ask your staff (open, in a team meeting, with a stated no-blame period for what has already happened):

1. Which AI tools have you used for work this month, including on your phone?
2. What did you put into them: client names, addresses, notes, voice recordings, screenshots of the planning?
3. What did the tool do for you that our current systems do not?
4. How did you check its answer before using it?
5. What would make it easier to tell me (an approved tool, training, a named person to ask)?

Approved AI can unsettle people too. At Vitalis the planning tool made visible the gaps in the roster, moments with no tasks, and staff felt that they had been doing work that was not there. Vitalis says leaders must state clearly that the tools are not meant to dismiss people or reorganise departments (Vilans, Vitalis interview, 2025). At Sint Anna Boxmeer, nurses, carers and quality nurses were involved from the start so everyone felt heard (Vilans, Wegwijzer, 2025).

If Thuiszorg Totaal has a works council, involve it early. Dutch law gives it consent rights over rules on processing employees' personal data, over systems able to monitor attendance, behaviour or performance, and over working-time arrangements (Wet op de ondernemingsraden, art. 27; Autoriteit Persoonsgegevens, n.d.). Whether a scheduling tool triggers them needs checking case by case.

3. Illustrative use case (CRISP-DM)

Linking with Chapter 1, Thuiszorg Totaal pilots a tool that suggests schedules and drafts client notifications.

(As with Chapter 1's example, Theme 4's own evidence search found no documented adopter case for automatic client-facing notifications — only vendor claims that failed the evidence gate. Read "drafts client notifications" below as an illustration of the trust rules, not as a use case the team currently recommends; Theme 4 recommends staff notifications instead, section 2.)

- **Business understanding:** the goal is less manual planning work, with the trust rules set before the pilot starts: what the tool may do (Part B) and who signs off.
- **Data understanding:** list every data field in three columns: hand over, keep, unsure. Unsure goes to the privacy officer.
- **Data preparation:** send only the minimum: availability, time slots, client codes and travel times calculated in-house.
- **Modelling:** the AI proposes schedules and drafts messages. The planner approves. Every suggestion is logged as accepted, changed or rejected.
- **Evaluation:** track how often planners override and why, and any wrong or missed visit notices. Check that no free-text client information appears in what is sent to the tool. Ask staff the five questions in Part C anonymously as well.
- **Deployment:** roll out gradually, with the written AI rule and training in place, a quarterly review, and the exit plan from Chapter 1.

4. Recommended tooling

Option	What it does for trust	Effort
One-page AI rule plus a list of approved tools	Tells staff what may never be entered and where to ask; gives them an alternative to free chatbots	Low: writing and a team meeting
Business AI workspace under a data processing agreement	Keeps use inside a contract: check data location, retention, logging and whether inputs are reused (Barberá, 2025)	Low to medium: vendor check and set-up
Decision log (spreadsheet or the tool's own log)	Shows whether "planner decides" is real; supports oversight duties	Low: agree fields and review monthly
Model on own or supplier-managed hardware	Maximum control over data	High: hardware, expertise, security; consider later (Dennstädt et al., 2025)

5. Responsible AI & trust

- **Does the data stay ours?** Only if the firm decides what is sent and checks the contract (Part A). Free tools give no such control.
- **How far can the output be relied on?** Language models can produce wrong statements that look convincing and can reflect bias (Dennstädt et al., 2025). Treat every suggestion as a draft, and test it in the pilot by counting errors and overrides. This chapter did not test any specific tool.
- **Keep Humans in the loop.** For critical uses, outputs should be reviewed by a person before use, and users should be taught not to over-rely on the model (Barberá, 2025). Oversight must be real (Part B).
- **What it means for the people doing the work.** Carers and planners should be told what the tool does with their data and schedules, have a clear way to correct a wrong schedule, and not be blamed for honest disclosure. The AI Act also requires workers to be informed before a high-risk workplace system is used (Article 26).
- **Limits.** Most sources here are about healthcare or workplaces in general. We found no study of AI scheduling in Dutch home care in this search.

6. First five steps for an owner

1. **Sort the data.** Make three lists with your privacy officer on what to hand over, keep, and where it is unsure.
2. **Write one page.** What AI staff may use for what, what never goes in, and who to ask. Offer an approved alternative.
3. **Ask your staff** the five questions in a team meeting. Agree your breach procedure first, and announce a no-blame period.
4. **Set the pilot rules.** The AI suggests, the planner decides, every override is logged, and you name in advance what would make you stop.

5. **Ask the vendor three questions in writing:** Where does our data go and is it kept or reused? Which AI Act class is the tool? What can we export and delete our data (Chapter 1)? Inform staff, and the works council if you have one, before the pilot starts.

7. Sources

Scores use the search criteria's 100-point rubric. (Relevance 25, Reliability 20, Evidence 30, Recency 10 and Context 15)

Sources	Rel. /25	Rel'y /20	Evid. /30	Rec. /10	Ctx /15	Total	Gate
S1 AP (2024)	22	20	18	5	15	80	pass
S2 Binnenlands Bestuur (2025a, b)	20	15	16	10	15	76	pass
S3 AP (2026)	18	20	24	10	15	87	pass
S4 AKD (n.d.)	20	16	16	5	15	72	pass
S5 Barberá (2025)	20	19	26	8	13	86	pass
S6 Dennstädt et al. (2025) †	18	17	18	8	9	70	pass
S7 European Commission, AI Act Service Desk	20	20	28	10	13	91	pass
S8 Council of the EU (2026); DLA Piper (2026)	15	18	24	10	12	79	pass
S9 Aza (2024)	20	20	26	8	11	85	pass
S10 Laux & Ruschemeier (2025)	18	17	24	8	11	78	pass
S11 Eurofound (2024)	12	18	20	8	8	66	pass
S12 Gillespie, Lockey et al. (2025)	20	17	22	8	6	73	pass
S13 AP (n.d.); WOR art. 27	18	18	26	5	15	82	pass
S14 Vilans, Vitalis interview (2025)	24	18	16	8	12	78	pass (narrowly)
S15 Vilans, Wegwijzer (2025)	22	18	18	10	13	81	pass

References

- S1 Autoriteit Persoonsgegevens. (2024). Let op: gebruik AI-chatbot kan leiden tot datalekken. The Dutch data protection regulator, reporting notifications it received itself.
- S2 Binnenlands Bestuur. (2025a, 30 December). 'AI-gebruik leidt tot meer datalekken'. Binnenlands Bestuur. (2025b, December). Eindhoven blokkeert ChatGPT na datalek. Dutch public-sector trade news relaying the regulator's warning and the municipality's own findings; secondary.
- S3 Autoriteit Persoonsgegevens. (2026). Datalek? Dit moet u doen. The regulator's own guidance on breach duties, up to date.
- S4 AKD. (n.d.). Het gebruik van AI-chatbots door zorgaanbieders: hoe datalekken te voorkomen? Dutch law firm writing for care providers; undated; used only for the professional-secrecy point.
- S5 Barberá, I. (2025). AI privacy risks & mitigations: Large language models (LLMs). European Data Protection Board, Support Pool of Experts. Commissioned by the EU data protection board, detailed and referenced; states that the views are the author's, not the EDPB's.
- S6 Dennstädt, F., Hastings, J., Putora, P. M., Schmerder, M., & Cihoric, N. (2025). Implementing large language models in healthcare while balancing control, collaboration, costs and security. npj Digital Medicine,

8, 143. Peer-reviewed journal, hospital and university authors; a comment piece, Swiss, and one author declares a commercial interest.

- S7 European Commission, AI Act Service Desk. Article 26 and Annex III. The EU institution's own publication of the Act; read as excerpts.
- S8 Council of the EU. (2026, 7 May). Artificial intelligence: Council and Parliament agree to simplify and streamline rules; DLA Piper. (2026). The Digital AI Omnibus: deferral of high-risk AI obligations. The Council states the agreed dates; the law firm confirms entry into force on 27 July 2026. Check the published regulation on EUR-Lex.
- S9 Aza, A. (2024). Scores as decisions? Article 22 GDPR and the judgment of the CJEU in SCHUFA Holding (Scoring) in the labour context. Industrial Law Journal, 53(4), 840–858. Peer-reviewed law journal, applies the ruling to work; read as abstract and excerpt.
- S10 Laux, J., & Ruschemeier, H. (2025). Automation bias in the AI Act. arXiv:2502.10036 (forthcoming in European Journal of Risk Regulation). Oxford and Hagen law scholars; a preprint, not the final version.
- S11 Eurofound. (2024, 22 November). Italy: Privacy authority fines Foodinho 5 million euro for rider data breaches. Platform Economy Database. EU agency database summarising a regulator's decision; gig delivery, not care.
- S12 Gillespie, N., Lockey, S., et al. (2025). Trust, attitudes and use of artificial intelligence: A global study 2025. University of Melbourne and KPMG. University-led survey of 48,000+ people, co-run with a consultancy; self-reported and global; only the media release was read.
- S13 Autoriteit Persoonsgegevens. (n.d.). Het OR-privacyboekje; Wet op de ondernemingsraden, art. 27 (confirm on wetten.overheid.nl). The regulator explaining works-council rights; the statute itself. Booklet undated.
- S14 Vilans. (2025, 9 July). Plannen en roosteren met AI bij Vitalis. Waardigheid en trots voor de toekomst. National care-knowledge institute, programme run with the Ministry of Health; first-hand interview with the organisation's innovation manager, no measured results, intramural care.
- S15 Vilans. (2025). Wegwijzer slimmer plannen (met AI): geleerde lessen uit de praktijk. Same publisher; built on interviews with 16 care organisations, published as short summaries.

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Theme 3 - SME knowledge gaps

Thuiszorg Totaal – AI for Scheduling and Notifications

Team 3 · version 2 · SME: Thuiszorg Totaal · 6 October 2026 · reason for this version: renumbered from Theme 4 to Theme 3; added provenance box, Plan B disclosure, the full gap statement, two reused citations, and the disclaimer footer

How this page was made. This page is team-authored scoping, not AI-drafted narrative: it records what the team already knows, what it still needs from the interview, and where the team's assumptions might be wrong — written from the research proposal and class discussion rather than from external literature. Evidence plan: Plan B. Thuiszorg Totaal has given a green light for an interview, but the visit has not yet happened (see Theme 4's evidence note). Where this page makes a legal or factual claim rather than stating a team assumption, it points to the appraised source already used in Chapter 1 or 2 rather than re-appraising it here. Check (not yet done): the team must confirm this page's gap statement still matches what Theme 4 assumes, and update both once the real interview happens.

1. Introduction

Our project focuses on Thuiszorg Totaal, a home-care company in Arnhem. We are researching how AI could help the company improve employee scheduling and automatic notifications.

Before recommending AI, we need to understand what the company already does, what information is missing, and whether AI would actually be useful.

2. What the Company Already Knows and Does

Thuiszorg Totaal provides home-care services, where employee schedules and communication are important for daily operations.

Our project focuses on how schedules are managed and how employees are informed about changes.

However, we still need to confirm which scheduling software the company uses, how much work is done manually, and what problems employees experience.

3. What the Company Says It Needs

Our team is investigating AI-supported scheduling and notifications as a possible solution.

However, we cannot yet confirm that Thuiszorg Totaal has specifically requested this solution. We need information from the company to understand its actual needs.

Possible needs include:

1. Reducing time spent creating schedules.
2. Informing employees faster about schedule changes.
3. Reducing scheduling mistakes.
4. Improving communication between employees.

These are possible needs, not confirmed findings from the company.

4. What Our Research Shows We Need to Know

Before deciding whether AI is suitable, we need answers to several questions.

Current software: What scheduling system does Thuiszorg Totaal currently use? Does it already offer automatic notifications?

Costs: How much would implementing an AI solution cost, and would the benefits justify the investment?

Privacy: How can employee and client information be protected under GDPR? (See Chapter 1's vendor-chain check and Chapter 2's hand-over/keep table — the answer depends on the vendor chosen, not on a fixed rule.)

Reliability: Can the system handle schedule changes correctly, and how would employees correct mistakes?

Employee acceptance: Would employees be comfortable using the system? (Chapter 2, section 2.2 already found that a global survey reports more than half of employees hide informal AI use at work (Gillespie, Lockey et al., 2025) — worth checking directly rather than assuming Thuiszorg Totaal's staff are the exception.)

We also need examples from comparable home-care organizations that have implemented similar systems successfully.

5. Most Important Knowledge Gap

The biggest knowledge gap, and the one this handbook's Theme 4 chapter is written to help answer, is:

> **Thuiszorg Totaal doesn't know whether automated notifications would actually reduce the workload described in the team's research proposal — or what a minimum version would cost in data, setup effort, and vendor risk — versus assuming any AI tool would simply handle it.**

A related, unanswered gap sits underneath it: whether Thuiszorg Totaal actually needs AI at all, or whether its existing scheduling software could already solve the problem. For example, if the company already has software that supports automatic notifications, introducing another AI tool might create unnecessary costs. We first need to understand the company's current system and identify which tasks still require manual work.

6. Difference Between Company Needs and Research Evidence

Our proposed solution focuses on AI automation.

However, this does not automatically mean that AI is the best solution.

The company may want faster scheduling and better communication, while our research may show that improving existing software or working processes would be more practical.

At this stage, this difference is a possibility rather than a confirmed finding. We need company interviews and supporting research to determine whether it exists.

7. Conclusion and what an owner can do now

Before recommending AI to Thuiszorg Totaal, we need to understand its current scheduling process, existing software, costs, privacy requirements and employee needs.

The most important question is whether AI would provide additional benefits compared with the company's existing tools.

Three concrete next steps, ahead of the real interview:

1. **Ask what scheduling software is already in place**, and whether it already does what a new AI tool would promise — before any vendor conversation starts.
2. **Ask the five staff questions from Chapter 2** informally, to get an early read on how much AI is already in use and how staff would react to a formal tool.
3. **Read Theme 4** alongside this page: it names the one use case (staff notifications) the team currently has real evidence for, so any interview question can test that specific idea rather than AI in general.

This will help the team make a recommendation based on evidence rather than assuming that AI is always the best solution.

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Theme 4 - AI use cases at SMEs, what do firms use AI for, and what pays off?

This chapter helps the owner of a small Dutch home-care organisation see what AI is actually being used for in care, not just promised, and whether one of those uses is worth trying.

Team 3 · version 2 · 6 October 2026 · sources last appraised 6 October 2026 · reason for this version: renumbered from Theme 5 to Theme 4 (content unchanged)

Evidence note for the reader. Thuiszorg Totaal has given a green light for an interview, but the visit has not yet happened. Everything on this page rests on desk research, with published reports, a sector whitepaper, and the front-runner hunt the team did in class, not on what Thuiszorg Totaal itself does or wants. Once the interview happens, real answers to "what have you tried or dropped" should replace the assumptions here. This page cannot yet tell the owner whether staff would actually want this, or what their current fall-response process costs them in time.

1. The question, for this kind of SME

What do home-care organisations like Thuiszorg Totaal actually use AI for today, and which of those uses has evidence behind it rather than a sales pitch? This chapter follows Theme 3's gap:

> **Thuiszorg Totaal doesn't know whether automated notifications would actually reduce the workload described in the team's research proposal — or what a minimum version would cost in data, setup effort, and vendor risk, versus assuming any AI tool would simply handle it.**

This page tries to answer that gap with one concrete, evidenced use.

2. The translation

Across the Dutch sector whitepaper, the Vilans case material already used in Chapter 2, and the team's front-runner hunt in class, the uses actually in operation cluster into five groups:

Use	What it does	Example found	Who it's for
Automatic staff notification on an event	A monitoring system detects an event (a fall, a missed check-in) and alerts the care team automatically	Zorg in Oktober, a Dutch care organisation: a system alerts staff within about a minute of a detected fall	Staff
Remote monitoring / night supervision	Replaces or reduces physical visits with a camera or sensor check	Swedish municipalities (Linköping, Söderköping, Gotland), evaluated by Socialstyrelsen	Clients, staff
Prediction from care notes	Flags clients likely to fall or deteriorate, from the notes carers already write	Cera (UK): predicts falls and deterioration risk from visit notes	Staff, planners
Administrative automation	Voice-to-text reporting, automatic scheduling, paperwork sent to the right system	Patyna (voice reporting), Sint Anna Boxmeer (AI scheduling, already covered in Chapter 2)	Staff
Client/family notifications	Automatic updates to clients or family after a visit	Found only as vendor claims (see below) — no documented adopter case	Clients, family

The finding this produces

The sector is automating what staff see and do — monitoring, prediction, scheduling, reporting — well ahead of automating what it tells the client or family. Every example of client-facing automated messaging we found was a software company's own product page, which your week 2 and week 3 rules already treat as a claim, not evidence. That gap is itself worth telling Thuiszorg Totaal: a tool that promises to "keep families updated automatically" is, on the evidence available to us, unproven in practice, however common the pitch is.

So this page carries staff notifications through the use case below, because it is the one with a real, named adopter behind it, and because it still answers the gap: an automatic alert replacing a phone call or a missed signal is the same kind of workload problem your team's proposal pointed to.

3. Illustrative use case (CRISP-DM): automatic staff notification on a care event

Business understanding. The goal is not to replace the planner's or carer's judgement, but to make sure a time-sensitive event — a fall, a missed scheduled check, a reading outside the agreed range — reaches a person who can act, without relying on someone happening to notice or call in. This directly follows Chapter 2's rule that the AI may act alone only for narrow, logged actions with a person always reachable (section 2.2 of Chapter 2).

Data understanding. What would trigger the alert depends entirely on what Thuiszorg Totaal actually monitors today — this is unknown and must come from the interview. Possible triggers, in rising order of data sensitivity: a missed scheduled visit confirmation (low sensitivity, operational data only); a sensor-based fall or inactivity signal (moderate, requires consent and a device in the home); a change flagged from care notes (high sensitivity, touches health data directly and falls under Chapter 2's "keep internal" column).

Data preparation. Following Chapter 2's table, only the trigger event and a client code should leave the building, never the underlying care note or a name. If the event is derived from care notes (as in Cera's model), that processing should happen inside the record system or a contracted, vendor-chain-checked tool (Chapter 1), not by sending notes to a general AI tool.

Modelling. A rule-based trigger ("alert if no check-in within X minutes of scheduled time") is simpler, cheaper, and easier to explain to staff and the AI Act than a predictive model trained on care notes. The Vilans & ActiZ whitepaper notes that most deployed examples in Dutch elderly care still combine rule-based logic with limited learning components, not a full predictive model — a smaller first step is both cheaper and closer to what is actually in use.

Evaluation. Before any rollout: how long after the real event does the alert arrive, how many false alerts fire, and whether staff report that it changed anything about how fast they responded. The Vilans & ActiZ whitepaper's own caution applies directly: sensor systems that alert too often cause "alarm fatigue," so the evaluation should explicitly ask staff whether they still act on every alert after a few weeks.

Deployment. Start with the single highest-value trigger agreed with staff (not necessarily falls — it should come from the interview), log every alert and every response, and review after a defined pilot period. Tell staff plainly, as Chapter 2 recommends, that the aim is faster response, not fewer staff.

4. Recommended tooling

Option	What it does	Effort
Rule-based alert on an existing system (a missed check-in, a set threshold)	The simplest version of the Zorg in Oktober pattern — no AI model needed yet	Low: configuration within an existing scheduling or sensor system
Sensor or wearable + alert routing	Adds a physical detection layer (fall, inactivity)	Medium: hardware, consent process, integration
Prediction from care notes (Cera-style)	Flags risk before an event happens, not just after	High: needs clean, structured care-note data and a vendor-chain check (Chapter 1)
Client/family notification layer	Automatic updates to clients or family	Not recommended yet — no adopter evidence found; treat any vendor promise here as unproven until a documented case exists

5. Responsible AI & trust

- **This still follows Chapter 2's rules.** An alert is at most "acts alone" for a narrow, logged action (section 2.2): it tells a person something happened, it does not decide care. Anything that changes what care is given stays "never alone."
- **Data stays minimal.** Only the trigger and a client code should travel outside the record system (Chapter 2, section 2.1's "hand over / keep" table applies here too).
- **Staff should not be alarmed by the data, the way Vitalis's staff were by visible gaps in the roster (Chapter 2).** A notification tool can surface how often something nearly went unnoticed, which can feel like criticism if introduced without explanation.

- **Limits of this page's evidence.** The strongest case (Zorg in Oktober) is a single named quote from a 2023 sector whitepaper, not an independent measurement, and it describes an elderly-care organisation, not a home-care one specifically. No adopter case was found for client-facing notifications at all. This page's own front-runner material (Cera, Sweden) is one step removed from a small Dutch home-care firm, as Chapter 2 already found for scheduling.

6. First five steps for an owner

1. **Ask staff, in the interview, what already goes unnoticed.** A missed check-in, a delayed response, a call that should have come sooner — the real trigger should come from them, not from a vendor's feature list.
2. **Start with the rule-based version, not a predictive model.** It is cheaper, easier to explain, and matches what is actually deployed in Dutch elderly care today.
3. **Decide what data the trigger needs, using Chapter 2's table**, before talking to any vendor.
4. **Set a false-alert tolerance before the pilot**, so "alarm fatigue" is caught early rather than after staff have already started ignoring alerts.
5. **Do not promise client or family notifications yet.** Say to staff and to clients that this is being tested for internal alerts first, because that is where the evidence currently is.

7. Sources

How sources were scored

Rubric: Relevance 25, Reliability 20, Evidence 30 (minimum 15 to pass), Recency 10, Context 15. All scores are AI-drafted and must be confirmed by the team.

#	Source	Rel. /25	Rel'y /20	Evid. /30	Rec. /10	Ctx /15	Total	Gate	Band
S1	Vilans & ActiZ whitepaper — Zorg in Oktober fall-alert case	25	20	24	6	14	89	pass	very good, flag recency
S2	Swedish municipalities, night supervision (Socialstyrelsen-evaluated)	15	24	26	7	10	82	pass	very good
S3	Cera (UK), front-runner card (multiple press sources)	20	15	18	10	8	71	pass	good — outcome % are company figures
—	Vendor pages for client/family notification apps (excluded)	20	10	6	9	13	58	fail	excluded — claims only, no documented case

References

- S1 Van Breda, L., Dircks, J., Hofstede, B., Nap, H. H., & Lukkien, D. (2023, November). AI in de ouderenzorg [Whitepaper]. Vilans & ActiZ. Published jointly by the national knowledge institute for care (Vilans, already used in Chapter 2) and the sector association for about 400 Dutch care organisations (ActiZ); the Zorg in Oktober example is a named, quoted practitioner (Robert Tournoij, zorginnovatie coach en ICT-medewerker). Weakest point: November 2023, about three years old — flagged on the page.
- S2 Socialstyrelsen (the Swedish government's health and welfare agency), Robotar och kameror kan göra omsorgen bättre (2 September 2025), plus municipal sources (Linköping, Söderköping, Lerum, Karlskoga, Gotland) and Sveriges Radio, as compiled on the team's front-runner card, 5 October 2026. Socialstyrelsen is an independent government evaluator, not a vendor or municipal self-report. [Team: confirm exact source list and URLs from your own card — these were reconstructed from the card text, not re-fetched.]
- S3 Compiled from: Cera conference abstract, ISPOR Europe 2024; TechCrunch, HLTH and Healthcare Technology Report (Cera's $150M raise); Healthcare Management and Government Technology (NHS rollout); MobiHealthNews (misconduct reports) — as compiled on the team's front-runner card, 5 October 2026. A mix of independent trade press and investor/company announcements. The hospitalisation and falls

percentages are Cera's own figures, not independently verified; the misconduct report is independent and should be weighed alongside the positive coverage.

- Excluded: Caring Village, ShiftCare, CareSmartz360, enCappture (English-language home-care communication apps); Octaaf, AIStudio Assist, AIFAIS, Red Factory, ProjectImpact.nl (Dutch care-software vendors) — all company pages describing a feature, with no named adopter and no measured outcome.

What remains open: the real Thuiszorg Totaal interview, which trigger staff would actually value, confirmation of every score above, and the exact source list for S2 and S3 from the team's own front-runner cards rather than this page's reconstruction of them.

Educational, not consultancy. This chapter offers inspiration and starting points, not legal advice or guarantees. Decisions about a firm remain the firm's own.

Tooling index

The accessible, mostly no-code tools referenced across the chapters, in one place: what each is for, what it costs in money and effort, and which chapter shows it in use.

Regional ecosystem & support

Where an owner in the region can go next: innovation centres, hubs, training, funding and networks. Organisations are named only with their agreement, and people are not named.

About this handbook

Who writes it. Students of HAN University of Applied Sciences, in teams, as the integrated task of the AIBS module (the research) and the AEL module (the platform it is researched and written with). Each cohort adds new chapters and refreshes existing ones, so the handbook compounds over time.

The quality bar. Every chapter is measured against the same four criteria before it is published: clarity for a non-technical reader, accuracy against appraised sources, responsible-AI coverage, and usefulness to an SME owner. Nothing reaches an SME reader unreviewed — chapters are curated by the course team before publication.

Educational, not consultancy. This handbook offers inspiration and starting points, not commercial advice or guarantees. Every chapter carries this disclaimer. Decisions about a firm remain the firm's own.

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